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Project Introduction

ParaliWatch

A Self-Updating Satellite Fire Tracker for Punjab's Paddy-Residue Burning Season.

- **Real-time monitoring** using NASA FIRMS VIIRS data
- **Data-driven insights** on daily and seasonal fire activity
- **Next-day forecasting** with Ridge Regression



The Inspiration

To turn raw satellite data into a simple, reliable, and continuous view of stubble burning in Punjab.

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The Problem

Paddy-Residue Burning



Every year, after the paddy harvest, large amounts of crop residue (***parali***) are often burned to quickly clear fields for the next wheat crop.

What is Paddy-Residue Burning?

The practice of setting fire to leftover paddy stalks (stubble) in the field after harvest.

Why is Monitoring Important?

- Track the **real scale and timing** of burning.
- Understand **spatial and seasonal patterns**.
- Support **evidence-based policy** and **sustainable alternatives**.

What Do Existing Raw Fire Maps Lack?

- Mostly show only raw hotspot locations on a map.
- No clear daily or seasonal trend analysis.
- No forecasting or performance tracking.



The Need »

A continuous, data-driven system that not only tracks fire activity but also analyses trends and helps anticipate the next day's risk.

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The Solution

ParaliWatch

A data-driven environmental intelligence platform that combines satellite observations, weather data, and machine learning to monitor and analyze paddy-residue fires in Punjab.

NASA FIRMS (VIIRS 375m)

- Near-real-time fire detections.
- Active thermal hotspots (FRP).
- Filtered for nominal & high confidence events.

Open-Meteo Weather Data

- Historical & daily weather records.
- Temperature, precipitation, wind and other covariates.
- Data from 5 monitoring hubs (Amritsar, Ludhiana, Patiala, Sangrur, Bathinda).

Interactive Fire Monitoring

- Geospatial map of detected fires across Punjab.
- Filter by date, pass (day/night), confidence and FRP intensity.
- Study region boundary and detailed hotspot insights.

Daily Trends & Analytics

- Daily fire activity and 7-day rolling trends.
- Seasonal cumulative analysis.
- A clear, data-backed view of the burning pattern.

The Outcome » *From raw satellite data to actionable insights — a unified platform to **monitor, analyze** and **understand** stubble burning in Punjab.*

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The Technology

Behind ParaliWatch

A modern, end-to-end data pipeline that brings together satellite data, weather intelligence, machine learning, and a scalable web platform.

Python & Pandas

- Data ingestion and processing.
- Create features and time-series datasets.
- Prepare data for forecasting and analysis.

Ridge Regression

- Next-day fire count forecasting on log(1 + Count).
- Uses lag, rolling and weather covariates.
- Benchmarked against Persistence and 7-Day Average.

GitHub Actions

- Automated daily data pipeline (02:37 UTC).
- Fetches latest satellite & weather data.
- Updates processed data and model outputs.

Next.js & Vercel

- Modern, responsive web interface (Next.js App Router).
- Interactive maps, charts and dashboards (Leaflet, Recharts).
- Deployed on Vercel for fast, reliable access.

End-to-End Workflow

Data Sources
NASA FIRMS + Open-Meteo

→

Data Processing
Clean, merge & feature engineering

→

ML Model
Ridge Regression (α = 1.5)

→

Automation
GitHub Actions (Daily Update)

→

Web Platform
Next.js on Vercel (Live Dashboard)

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What We Built

A complete, data-driven platform to monitor, analyze, and forecast paddy-residue fires in Punjab.



Key Features

Satellite Monitoring
Real-time fire detections (NASA FIRMS VIIRS).

Interactive Fire Map
Spatial view with filters and FRP intensity.

Trends & Analytics
Daily activity and seasonal patterns.

Weather Intelligence
Temperature, wind, precipitation across 5 monitoring hubs.

ML Forecasting
Next-day estimate using Ridge Regression.

Automated Pipeline
Daily data update via GitHub Actions.

Project Outcome

- A working end-to-end environmental intelligence platform.
- Combines satellite, weather and machine learning in one place.
- A reproducible and self-updating system for monitoring stubble fires.

GitHub Repository
github.com/dk-khandelwal06/paraliwatch

Live Website
paraliwatch.vercel.app